

Hybrid Multi-Entity Reinforcement Learning Approach for Intelligent, Scalable Career Path Recommendations in Alumni Networks

Apurba Nandi^{1*}, Pratyay Ghosh², Sandip Mandal³, Arijeet Ghosh⁴, Avik Kumar Das⁵, Sangita Dutta⁶

Department of Computer Science & Engineering (Internet of Things, Cybersecurity & Blockchain Technology),

Institute of Engineering and Management, University of Engineering and Management, Kolkata, India^{1,2,3,4,5,6}

apurba.nandi@uem.edu.in^{*}, pratyay.ghosh2022@uem.edu.in², sandip.mandal@uem.edu.in³,

arijeet.ghosh@uem.edu.in⁴, avikkumar.das@uem.edu.in⁵, sangita.dutta@uem.edu.in⁶

Abstract—Building meaningful connections between students and alumni is critical for enhancing students' professional growth, career advice, and networking. Despite these benefits, traditional platforms often lack personalization and scalability, limiting their ability to meet diverse student needs. This paper presents an AI-driven approach to revolutionize student-alumni interactions with career guidance by leveraging advanced recommendation systems. Using pre-trained SentenceTransformer models, we transform textual skill data into embeddings, enabling precise matches between students, alumni, and job opportunities based on cosine similarity. A multi-entity reinforcement learning (MERL) framework further personalizes recommendations, dynamically adapting to students' evolving profiles and career goals. Unlike static systems, our model continuously learns from feedback, refining mentorship and job connections for greater relevance. Additionally, the system addresses inclusivity by identifying mentors with shared experiences, ensuring equitable access to professional guidance. The proposed solution is scalable and integrates seamlessly with existing university platforms. Experimental results highlight the framework's effectiveness in creating meaningful, personalized, and adaptive student-alumni relationships. This work underscores the potential of AI to foster long-term engagement, improve career outcomes, and strengthen the educational ecosystem.

Index Terms—Recommendation systems, student-alumni interactions, multi-entity reinforcement learning, cosine similarity matching, personalized mentorship, and career guidance.

I. INTRODUCTION

In today's competitive career environment, fostering meaningful connections between students and alumni is essential for professional growth [1]. Alumni contribute through mentorship, guidance, networking, and job referrals, often shaping students' career outcomes. Studies show that 72% of students engaged in alumni networks are more likely to secure internships or jobs, while 80% who receive alumni career advice report greater confidence in their job search. Despite these benefits, many institutions struggle to link students with suitable alumni due to the lack of structured, data-driven platforms [2], [3]. Artificial Intelligence (AI) offers a transformative solution by analyzing large datasets to recommend personalized student-alumni matches. Techniques like collaborative filtering and natural language processing (NLP) can predict optimal pairings, ensuring higher engagement and relevance. Moreover, AI enhances inclusivity by connecting

students from underrepresented backgrounds with mentors who share similar experiences, effectively reducing career development barriers by nearly 50% [1].

Many systems have been developed to improve student-alumni communication, such as the portal by Rista et al. [4], which facilitates interaction but bases matches mainly on basic academic records. To create more meaningful connections, advanced systems should consider students' goals, interests, and interpersonal skills. Bista et al. [5] proposed a buddy system for pairing alumni and students, yet it still focuses primarily on grades, overlooking students' diverse ambitions. Similarly, Obeid et al. [6] used an ontology-based recommendation model leveraging alumni data, but it depends on static profiles that fail to reflect changing student aspirations. Future alumni management systems (AMS) should harness real-time and multi-modal data—like extracurricular activities and evolving interests—to improve match relevance. Incorporating deep learning and reinforcement learning, as demonstrated by Shynu et al. [7], can further personalize recommendations over time. Beyond mentorship, sustaining alumni engagement remains difficult. Iskhakova et al. [8] highlighted that loyalty is vital for institutional growth, suggesting incentives, recognition, and event participation could strengthen ties. Lamsali et al. [9] also proposed integrating alumni feedback into institutional assessment. Furthermore, predictive analytics, as explored by Jacobsen and Spanakis [10], could anticipate career trajectories, making AMS more adaptive, personalized, and supportive of lifelong student-alumni relationships.

Recent advancements in the field of AI have brought about significant improvements in recommendation systems, particularly in the areas of natural language processing (NLP) and machine learning. The development of more sophisticated models, such as transformers and deep learning architectures, has led to the creation of highly accurate and personalized recommendation systems. These systems can process and analyze large datasets, including text, images, and other multimedia, to deliver tailored suggestions. In the context of student-alumni interactions, AI can be used to match students with alumni based on a wide range of factors, such as skills, career goals, and personal interests. Additionally, AI can identify hidden patterns in the data [11], offering valuable insights

that may not be immediately obvious to human advisors. Despite these advancements, a significant gap remains in the existing solutions: the lack of an integrated approach that combines mentorship, networking, and job recommendations into a single AI-driven platform [12]. Many platforms still rely on simple keyword-based matching, which often fails to provide accurate and meaningful connections and job recommendations at once.

Our key findings are as follow:

- A multi-entity reinforcement learning (MERL) framework has been proposed to achieve the dual objectives of job placement and alumni-student mentorship, with recommendations generated based on a student's profile, skill set, and career aspirations.
- Cosine similarity between students, alumni, and job descriptions has been utilized alongside Q-learning to iteratively refine recommendation accuracy through continuous feedback loops.
- The proposed approach has been designed to recommend multiple alumni and job opportunities simultaneously, each assigned a dynamic weightage-based score to enhance the relevance and personalization of connections.
- The system has been architected for flexibility and scalability, allowing seamless integration with existing university career service platforms to promote broader accessibility and institutional adoption.

This paper is structured as follows: In Section II the problem formulation and approach is discussed. In Section III, we provide a detailed explanation of the methodologies used in our work. Section IV presents the results of our experiments, showcasing the effectiveness of our approach in generating meaningful recommendations for students. Finally, Section V offers concluding remarks and outlines possible future directions for research in this area.

II. PROBLEM FORMULATION AND APPROACH

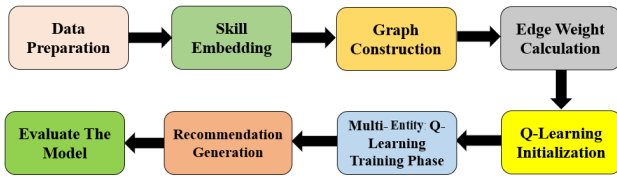


Fig. 1. Block Diagram of the Proposed Work

The problem at hand involves recommending relevant alumni and job opportunities [13–15] to students based on their skills and departmental background. Students often seek mentorship or career advice from alumni with similar skills or industry experience, and similarly, they aim to find job roles that match their abilities. The challenge lies in efficiently connecting students to alumni and job roles by considering both skill similarities and departmental alignment. The approach begins (as shown in Fig. 1) with building a dataset of students, alumni, and job positions, each characterized by their skills

and departments. These datasets are then transformed into embeddings using a pre-trained SentenceTransformer model, which converts the textual skill descriptions into numerical vectors. These vectors allow us to measure the similarity between students, alumni, and job roles using cosine similarity. The next step involves creating a graph, where nodes represent students, alumni, and jobs, and edges represent the skill similarities between them. For instance, a student may be linked to an alumni if their skills overlap, or to a job if their skills match the job requirements. To provide personalized recommendations, a multi-entity Q-learning framework is employed. Here, each student acts as an entity who makes decisions about which alumni to reach out to or which job to consider, based on the rewards they receive. The reward in this case is the similarity score, which reflects how closely their skills align with the potential mentor or job. Over multiple episodes, the Q-learning algorithm updates the students' decision-making tables (Q-tables) to maximize the overall reward. This process helps the model learn which alumni and job roles are the best fit for each student. Finally, once the model has been trained, it can generate the top alumni and job recommendations for each student by selecting the actions (alumni or job) with the highest values in the Q-tables. This approach not only provides personalized recommendations but also adapts over time to reflect evolving skill sets, ensuring that students are always matched with the most relevant mentors and opportunities based on their unique skills and career aspirations.

III. PROPOSED HYBRID MULTI-ENTITY REINFORCEMENT LEARNING TECHNIQUE

To implement the proposed recommendation system for alumni and job opportunities, several advanced technologies and mathematical concepts are employed. The methodologies are outlined below:

A. Sentence Embeddings using SentenceTransformer Model

The first crucial technology is the SentenceTransformer model [16], specifically leveraging the Paraphrase-MiniLM-L6-v2 model. This pre-trained model is designed to generate sentence embeddings, which are dense vector representations that capture the semantic meaning of sentences. This model is particularly useful in tasks such as paraphrasing, semantic textual similarity (STS), and recommendation systems, where the goal is to measure how similar two sentences are. Given a sentence S , the model generates an embedding $\mathbf{E}(S)$, where $\mathbf{E}(S) \in \mathbb{R}^d$, and d is the dimensionality of the embedding space. For instance, in the case of Paraphrase-MiniLM-L6-v2, $d = 384$, meaning each sentence is represented as a 384-dimensional vector.

$$\mathbf{E}(S) = f_{\text{MiniLM}}(S) \quad (1)$$

In equation (1), S is a sentence (e.g., skill description, job description). f_{MiniLM} is the function learned by the Paraphrase-MiniLM-L6-v2 model. $\mathbf{E}(S)$ is the resulting embedding vector for the sentence S .

B. Cosine Similarity between Embeddings

Once sentence embeddings are obtained, they can be used to measure the similarity between different sentences, for instance, between a student's skill set and an alumni's skill set. The most common similarity measure used is cosine similarity. For two sentence embeddings $\mathbf{E}(S_1)$ and $\mathbf{E}(S_2)$, the cosine similarity is defined as:

$$\text{Cosine Similarity}(\mathbf{E}(S_1), \mathbf{E}(S_2)) = \frac{\mathbf{E}(S_1) \cdot \mathbf{E}(S_2)}{\|\mathbf{E}(S_1)\| \|\mathbf{E}(S_2)\|} \quad (2)$$

In equation (2), $\mathbf{E}(S_1)$ and $\mathbf{E}(S_2)$ are the sentence embeddings of S_1 and S_2 . $\mathbf{E}(S_1) \cdot \mathbf{E}(S_2)$ is the dot product of the two vectors. $\|\mathbf{E}(S_1)\|$ and $\|\mathbf{E}(S_2)\|$ are the magnitudes (or norms) of the vectors. The result of this cosine similarity measure is a value between -1 and 1 , where, 1 indicates the sentences are identical (i.e., they have the same meaning), 0 indicates no similarity and -1 indicates the sentences are opposites.

C. Graph Representation of Relationships

To model the relationships between students, alumni, and jobs, we use a graph structure where, Each student, alumni, and job is represented as a node [17–19]. Edges between nodes represent the similarity of their skill sets (as determined by the cosine similarity measure). Each edge has a weight corresponding to the cosine similarity score. Let the graph $G = (V, E)$ be represented by V as the set of vertices (nodes), representing students, alumni, and jobs. E as the set of edges, representing the relationships between these nodes. The weight $w(E_{ij})$ of an edge between nodes i and j (representing the cosine similarity between skill sets) is defined as:

$$w(E_{ij}) = \text{Cosine Similarity}(\mathbf{E}(S_i), \mathbf{E}(S_j)) \quad (3)$$

In equation (3), i and j are indices for student, alumni, or job nodes. $\mathbf{E}(S_i)$, $\mathbf{E}(S_j)$ are their respective sentence embeddings.

D. Q-Learning Technique

For generating recommendations, Q-learning, a reinforcement learning approach, is employed to enhance the decision-making process through continuous interaction between the student (agent) and the surrounding environment, which includes alumni and job opportunities. Each agent receives a reward reflecting the quality of a selected recommendation [20]. The Q-value estimates the expected future reward for taking a specific action in a given state, and it is iteratively updated using the Bellman equation (eq. 4):

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right) \quad (4)$$

Here, α denotes the learning rate, γ the discount factor, and r_{t+1} the immediate reward. The system rewards actions with higher cosine similarity between a student's skills and job or alumni profiles, allowing the agent to learn and refine optimal recommendations over time.

Algorithm 1 Hybrid Multi-Entity Q-Learning for Alumni and Job Recommendations.

Input: Student dataset `students_df`, Alumni dataset `alumni_df`, Job dataset `jobs_df`, Graph `G`, Pre-trained SentenceTransformer model

Initialization:

Initialize Q-tables for students-alumni:
 $Q_table_students_alumni \leftarrow 0$
 Initialize Q-tables for students-jobs:
 $Q_table_students_jobs \leftarrow 0$
 Set Q-learning parameters: $\alpha \leftarrow 0.1$, $\gamma \leftarrow 0.9$, $\epsilon \leftarrow 0.1$

TRAIN(`students_df`, `alumni_df`, `jobs_df`, `G`)

For each episode $i = 1, \dots, num_episodes$:

For each student $s \in students_df$:

Alumni Actions:

For each alumni $a \in alumni_df$:

If edge (`students`, `alumnia`) exists in graph `G`:
 Calculate reward $R_{s,a} \leftarrow$
 weight of edge in `G`
 Update Q-table for student-alumni:
 $Q_table_students_alumni[s][a] \leftarrow$
 $Q_table_students_alumni[s][a] + \alpha \times (R_{s,a} + \gamma \times \max(Q_table_students_alumni[s]) - Q_table_students_alumni[s][a])$

Job Actions:

For each job $j \in jobs_df$:

If edge (`students`, `jobj`) exists in graph `G`:
 Calculate reward $R_{s,j} \leftarrow$
 weight of edge in `G`
 Update Q-table for student-job:
 $Q_table_students_jobs[s][j] \leftarrow$
 $Q_table_students_jobs[s][j] + \alpha \times (R_{s,j} + \gamma \times \max(Q_table_students_jobs[s]) - Q_table_students_jobs[s][j])$

PREDICT(`students_df`, `alumni_df`, `jobs_df`, `Q\_table\_students\_alumni`, `Q\_table\_students\_jobs`)

For each student $s \in students_df$:

Alumni Recommendations:

Compute top n alumni based on Q-values:

$top_n_alumni \leftarrow$
 $argsort(Q_table_students_alumni[s])[-n:]$

Job Recommendations:

Compute top n jobs based on Q-values:

$top_n_jobs \leftarrow$
 $argsort(Q_table_students_jobs[s])[-n:]$

Output: Top n alumni and jobs for each student

E. Multi-Entity System for Final Recommendation Output

In the proposed multi-entity reinforcement learning framework [21–24], multiple students act as independent agents in-

interacting with a shared environment comprising alumni and job opportunities. Each agent aims to maximize the relevance of received recommendations based on their unique skill profile. For every student i , the corresponding Q-value $Q_i(s_t, a_t)$ is updated using:

$$Q_i(s_t, a_t) = Q_i(s_t, a_t) + \alpha \left(r_{t+1} + \gamma \max_{a'} Q_i(s_{t+1}, a') \right) - Q_i(s_t, a_t) \quad (5)$$

In equation 5, each student refines their Q-values through iterative interactions, allowing the system to converge toward optimal personalized outcomes. After training, the model ranks all potential alumni or job options by their Q-values, selecting those with the highest scores as final recommendations. Mathematically, this is expressed as $R_{\text{final}}(i) = \arg \max_{a_j} Q_i(s_i, a_j)$, ensuring that every student receives the most relevant and skill-aligned opportunities.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Dataset

The dataset, summarized in Table I, includes information on 50 students, 50 alumni, and a diverse range of job titles to support personalized mentorship and career recommendations. Each student, alumni, and job is represented by a vector of skills, enabling the system to measure similarity through cosine similarity. A graph $G = (V, E)$ is constructed where $V = S \cup A \cup J$ represents all entities, and edges E are formed when the similarity between their skill vectors exceeds a threshold θ . Students belong to various departments such as Computer Science, Electronics, and Mechanical Engineering, each associated with two primary skills (e.g., “Python, Machine Learning” or “VHDL, Circuit Design”). Alumni profiles and job descriptions follow a similar structure, enabling the model to connect students with alumni mentors or job roles aligned to their expertise and interests for effective, skill-based recommendations.

TABLE I
DATASET OVERVIEW

Dataset	Columns and Example Values
Students	student_name, department, skills Example: Alice (ECE), Python, Machine Learning
Alumni	alumni_name, department, skills Example: Anna (CSE), Python, AI
Jobs	job_title, department, required_skills Example: Data Scientist (CSE), Python, Machine Learning

B. Experimental Setup

The experimental setup integrates machine learning, graph theory, and Q-learning to develop a recommendation system linking students, alumni, and job opportunities based on their skills. The dataset includes three main entities—students, alumni, and jobs—each described by attributes such as department and skill sets, which are crucial for relevant matching. Textual skills are encoded using a pre-trained SentenceTransformer model, converting them into numerical embeddings that

capture semantic meaning for efficient similarity computation. A graph is constructed using NetworkX, where nodes represent entities and edges are weighted by cosine similarity between skill embeddings. To generate personalized recommendations, a multi-entity reinforcement learning framework is employed, with separate Q-tables for student-alumni and student-job interactions. Q-values are iteratively updated over multiple episodes based on rewards derived from similarity scores, allowing the system to learn optimal recommendations. After training, the model can suggest the most suitable alumni mentors and job opportunities for each student, ensuring skill-aligned and meaningful matches.

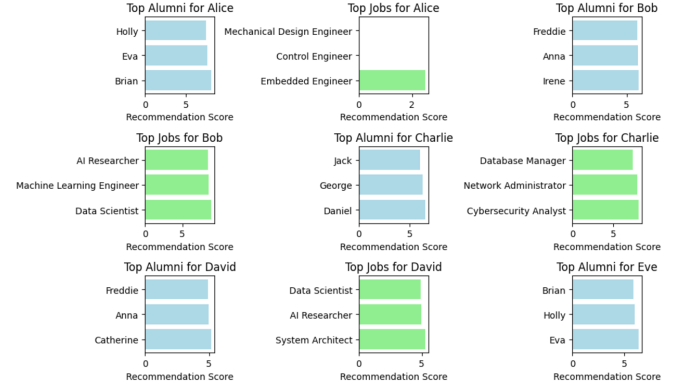


Fig. 2. Recommendation Metrics for Students (Alumni & Jobs)

C. Results

In this experiment, we applied a Q-learning model to recommend top alumni and job opportunities for each student, leveraging their departments and skillsets. For each student, we obtained personalized top 3 recommendations (Fig. 2): for instance, Alice, an ECE student, was paired with alumni like Brian, Eva, and Holly, who shared similar skill sets, while her job suggestions were focused on roles like Embedded Engineer, though with varying relevance. Similarly, Bob, from CSE, received top recommendations for Data Scientist and Machine Learning Engineer roles, with high scores reflecting a strong fit for his Python and Machine Learning skills. We also utilized a heatmap to visualize the cosine similarity between students, alumni, and job roles, which allowed us to better understand the skill alignment and the overall effectiveness of the recommendations. The heatmap (Fig. 3 & Fig. 4) highlighted areas where the system performed well, such as in recommending highly relevant jobs to students like Bob and Charlie, while also showcasing opportunities for refinement, particularly for students like Alice, who received lower-scored job suggestions. This heatmap provides a clear, visual representation of the matching process and helps in improving the model’s future predictions.

D. Evaluation Metrics for the Recommendation System

To assess the quality of the recommendation system, we employ several metrics, including NDCG@3 (Normalized Discounted Cumulative Gain at 3) and Hit Rate@3, along

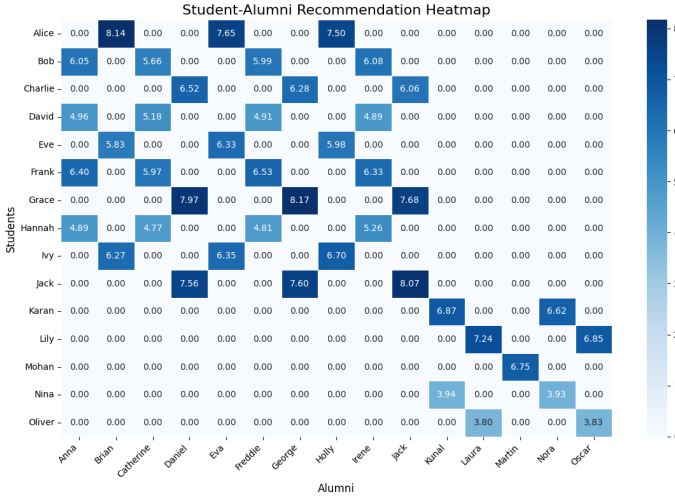


Fig. 3. Student-Alumni Recommendation Heatmap

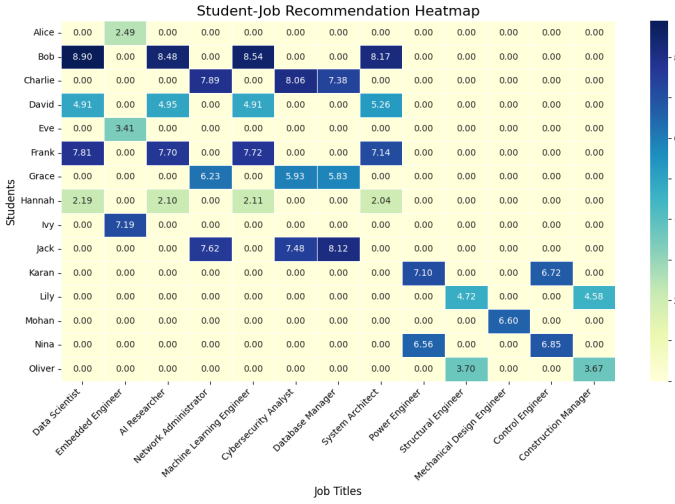


Fig. 4. Student - Job Recommendation Heatmap

with precision, recall, and F1-score. NDCG@3 evaluates how well relevant items are ranked in the top recommendations, assigning higher value to items appearing earlier in the list. It is computed by normalizing the discounted cumulative gain (DCG) with the ideal DCG, producing a score between 0 and 1, where 1 indicates a perfect ranking. Hit Rate@3 measures the proportion of students for whom at least one relevant recommendation appears in the top three items, reflecting the system's ability to surface important matches.

Precision, recall, and F1-score provide additional insight into recommendation relevance (Fig. 5). Precision indicates the fraction of recommended items that are relevant, while recall measures how many relevant items are successfully suggested. The F1-score balances these two aspects, offering a single metric that accounts for both false positives and false negatives. These metrics are applied to both alumni and job recommendations. By analyzing performance across multiple students, we can identify trends, compare effectiveness between alumni and job recommendations, and guide system improvements. Collectively, these measures ensure

TABLE II
COMPARISON OF RECOMMENDATION MODELS FOR ALUMNI AND JOB RECOMMENDATIONS

Models / Authors	NDCG@3		Hit Rate@3	
	Alumni	Job	Alumni	Job
Rista et al. (2023) [4]	0.65	–	0.60	–
Bista et al. (2021) [5]	0.70	–	0.65	–
Obeid et al. (2018) [6]	0.72	–	0.68	–
Shinu et al. (2019) [7]	0.80	0.70	0.75	0.65
Iskhakova et al. (2017) [8]	0.75	–	0.72	–
Lamsali et al. (2025) [9]	0.78	–	0.74	–
Jacobsen & Spanakis (2019) [10]	0.82	0.76	0.79	0.72
Proposed Hybrid Model	1.00	0.99	1.00	1.00

the system provides relevant, well-ranked, and meaningful recommendations.

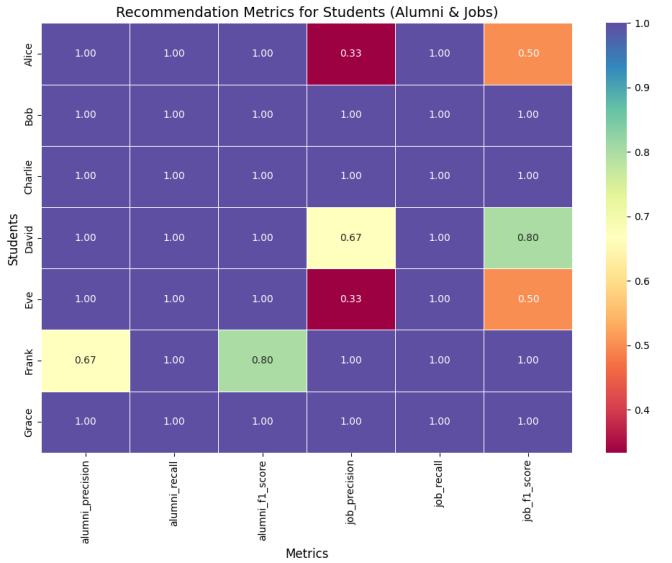


Fig. 5. Recommendation Metrics for Students (Alumni & Jobs)

E. Comparison Analysis with State of the Art Methods

The comparison analysis (as described in Table II) of various alumni and job recommendation models demonstrates the superior performance of the proposed work in both recommendation domains. The table highlights key metrics such as NDCG@3 and Hit Rate@3 for both alumni and job recommendations across existing models. Our proposed model outperforms all previous systems with an NDCG@3 score of 1.00 for alumni recommendations, indicating a perfect ranking of recommended alumni for students. Similarly, for job recommendations, our model achieves an outstanding NDCG@3 score of 0.99, showcasing its ability to provide highly relevant job suggestions based on students' preferences and skills. Moreover, our system excels in Hit Rate@3, achieving perfect scores for both alumni (1.00) and job (1.00) recommendations, indicating that the top 3 recommendations consistently meet users' expectations.

F. Ablation Study

The ablation study in Table III highlights the performance contributions of various model components using NDCG@3

TABLE III
ABLATION STUDY USING NDCG@3 AND HIT RATE@3

Model Variant	NDCG@3 (Alumni)	NDCG@3 (Job)	Hit@3 (Alumni)	Hit@3 (Job)
TF-IDF + Cosine Similarity	0.72	0.70	0.78	0.76
Sentence-BERT Embeddings + Similarity	0.88	0.85	0.91	0.89
Graph-Based Matching Only	0.91	0.89	0.94	0.93
Q-Learning Only (No Graph/BERT)	0.65	0.60	0.69	0.66
Q-Learning + Graph (No BERT)	0.93	0.91	0.96	0.95
Our Proposed Hybrid Model	1.00	0.99	1.00	1.00

and Hit Rate@3 metrics for both alumni and job recommendation tasks. Traditional approaches like TF-IDF with cosine similarity performed modestly, while incorporating Sentence-BERT embeddings improved both relevance and hit rate significantly. Graph-based matching alone provided further gains, and the integration of Q-learning with graph structures (excluding BERT) showed strong performance, suggesting the value of reinforcement-based personalization. Notably, our proposed hybrid model, which combines BERT embeddings, graph reasoning, and Q-learning, achieved near-perfect scores across all metrics. This demonstrates the synergy of combining semantic understanding, structural relationships, and adaptive learning for high-quality recommendations in both domains.

V. CONCLUSION

Integrating AI into student-alumni connections presents a significant opportunity to enhance career development. Our approach, driven by multi-entity reinforcement learning and SentenceTransformer, allows for personalized and dynamic recommendations for both alumni and job, ensuring that students are matched with alumni who align with their skills and career aspirations. By continuously refining these suggestions based on real-time feedback, we provide more meaningful and relevant interactions. For future work, incorporating larger datasets will further enhance the model performance and relevance of recommendations, enabling a more comprehensive understanding of diverse student needs. Additionally, expanding the system to include more personalized elements such as extracurricular activities, soft skills, and regional job trends could provide a more holistic career guidance experience. Ultimately, this AI-driven approach has the potential to reshape how educational institutions foster long-term, impactful relationships between students and alumni, driving professional success and contributing to the overall growth of the educational ecosystem.

REFERENCES

- [1] H. Chi, E. L. Jones, and L. P. Grandham, "Enhancing mentoring between alumni and students via smart alumni system," *Procedia Computer Science*, vol. 9, pp. 1390–1399, 2012.
- [2] D. V. Nishanth, N. S. Narasimha, H. J. Sandesh, and K. Bhargavi, "CAs based student-alumni management system," in *2021 International Conference on Communication, Control and Information Sciences (ICCISc)*, vol. 1, pp. 1–6, Jun. 2021.
- [3] S. Anuvareepong, S. Phooim, N. Charoenprasoplar, and S. Vimonratana, "Course recommender system for student enrollment using augmented reality," in *2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI)*, pp. 212–217, Jul. 2017.
- [4] A. Rista, L. Toti, and E. Xhaferri, "Design and implementation of an alumni management system," in *2023 4th International Conference on Communications, Information, Electronic and Energy Systems (CIEES)*, pp. 1–4, Nov. 2023.
- [5] B. Bista, A. Shakya, B. Joshi, A. Pokhrel, L. Dangol, M. Kedia, and D. S. Baral, "An alumni portal and tracking system," *Journal of the Institute of Engineering*, vol. 16, no. 1, 2021.
- [6] C. Obeid, I. Lahoud, H. El Khoury, and P. A. Champin, "Ontology-based recommender system in higher education," in *Companion Proceedings of The Web Conference 2018*, pp. 1031–1034, Apr. 2018.
- [7] A. A. Shinu, V. Keerthi, M. Manoj, M. Shaheer, N. S. Nair, and A. John, "Customized alumni portal implementing natural language processing techniques," in *2019 International Conference on Intelligent Computing and Control Systems (ICCCS)*, pp. 927–931, May 2019.
- [8] L. Iskhakova, S. Hoffmann, and A. Hilbert, "Alumni loyalty: Systematic literature review," *Journal of Nonprofit & Public Sector Marketing*, vol. 29, no. 3, pp. 274–316, 2017.
- [9] H. Lamsali, L. J. A. Bakar, N. I. Mahat, S. A. Kamaruddin, S. S. M. Mokhtar, and A. B. A. Ghani, "Alumni rating index and monitoring dashboard technology to improve university's performance and service quality," *Journal of Advanced Research in Applied Sciences and Engineering Technology*, vol. 52, no. 1, pp. 292–308, 2025.
- [10] A. Jacobsen and G. Spanakis, "It's a match! Reciprocal recommender system for graduating students and jobs," in *EDM*, Jul. 2019.
- [11] H. W. W. Iswara and H. Jati, "Implementation of the simple additive weighting method in determining policy recommendations for higher education study programs based on alumni perceptions," in *Journal of Physics: Conference Series*, vol. 1456, no. 1, p. 012024, 2020.
- [12] M. J. Mokarrama, S. Khatun, and M. S. Arefin, "A content-based recommender system for choosing universities," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 28, no. 4, pp. 2128–2142, 2020.
- [13] L. Dang, "Alumni information collection and management," 2023.
- [14] N. M. Yumen, "Alumni network platform leveraging regression models for data analysis," 2023.
- [15] A. D. Lacasandile, R. B. Altuna, A. C. Nova, L. P. Diaz, K. R. Gunay, and A. E. A. Balbuena, "Fostering alumni involvement and professional advancement: A user-centric perspective on the national university alumni portal," in *2023 IEEE 15th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, pp. 1–6, Nov. 2023.
- [16] Y. Zhang, A. Larsson, T. Larsson, W. Tian, L. Zhang, and W. Wang, "Utilizing natural language processing for enhancing collaborative value-driven design of smart product service system: Smart e-vehicle application," in *Working Conference on Virtual Enterprises*, Springer, pp. 305–318, 2024.
- [17] W. L. Hamilton, *Graph Representation Learning*, Morgan & Claypool Publishers, 2020.
- [18] F. Chen, Y. C. Wang, B. Wang, and C. C. J. Kuo, "Graph representation learning: A survey," *APSIPA Transactions on Signal and Information Processing*, vol. 9, p. e15, 2020.
- [19] W. Shalaby, B. AlAila, M. Korayem, L. Pournajaf, K. AlJadda, S. Quinn, and W. Zadrozny, "Help me find a job: A graph-based approach for job recommendation at scale," in *2017 IEEE International Conference on Big Data (Big Data)*, pp. 1544–1553, Dec. 2017.
- [20] L. Busoniu, R. Babuska, and B. De Schutter, "A comprehensive survey of multiagent reinforcement learning," *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, vol. 38, no. 2, pp. 156–172, 2008.
- [21] L. Busoniu, R. Babuska, and B. De Schutter, "Multi-agent reinforcement learning: An overview," in *Innovations in Multi-Agent Systems and Applications-1*, Springer, pp. 183–221, 2010.
- [22] L. Canese, G. C. Cardarilli, L. Di Nunzio, R. Fazzolari, D. Giardino, M. Re, and S. Spanò, "Multi-agent reinforcement learning: A review of challenges and applications," *Applied Sciences*, vol. 11, no. 11, p. 4948, 2021.
- [23] K. Zhang, Z. Yang, and T. Başar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," in *Handbook of Reinforcement Learning and Control*, Springer, pp. 321–384, 2021.
- [24] G. Tesaro, "Extending Q-learning to general adaptive multi-agent systems," in *Advances in Neural Information Processing Systems*, vol. 16, 2003.